

OS-PRACTICAL

Practical-6

6. Create a client-server application using Named Pipes (FIFOs). The client sends messages and the server processes and responds to them. Analyze FIFO behavior when multiple clients communicate simultaneously.

Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

```
ananya@DESKTOP-42I5F93:~$ pwd
/home/ananya
ananya@DESKTOP-42I5F93:~$ mkdir practical6
mkdir: practical6: File exists
ananya@DESKTOP-42I5F93:~$ cd practical6
ananya@DESKTOP-42I5F93:~/practical6$ pwd
/home/ananya/practical6
ananya@DESKTOP-42I5F93:~/practical6$ nano server.c
ananya@DESKTOP-42I5F93:~/practical6$ nano client.c
ananya@DESKTOP-42I5F93:~/practical6$ ls
client.c  server.c
ananya@DESKTOP-42I5F93:~/practical6$ gcc server.c -o server
ananya@DESKTOP-42I5F93:~/practical6$ gcc client.c -o client
ananya@DESKTOP-42I5F93:~/practical6$ ls
client  client.c  server  server.c
ananya@DESKTOP-42I5F93:~/practical6$ ./server
Server started...
Waiting for client message...
Client says: Hello Server
Response sent to client.
```

```
ananya@DESKTOP-42I5F93: ~ × + ▾
ananya@DESKTOP-42I5F93:~$ cd practical6
ananya@DESKTOP-42I5F93:~/practical6$ ls
client  client.c  client_to_server  server  server.c  server_to_client
ananya@DESKTOP-42I5F93:~/practical6$ ./client
Enter message for server: Hello Server
Server response: Server received: Hello Server
ananya@DESKTOP-42I5F93:~/practical6$ |
```



```
ananya@DESKTOP-42I5F93:~$ kill -SIGUSR1 651
```

```
ananya@DESKTOP-42I5F93:~$ ./signal
```

```
Process ID: 742
```

```
Signal handling program started...
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```
Program is running...
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Program is running...
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Program is running...
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```
SIGTERM received!
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```
Termination signal received.
```

```
Program is running...
```

```
ananya@DESKTOP-42I5F93:~$ kill -SIGTERM 742
```

```
ananya@DESKTOP-42I5F93:~$ ./signal
```

```
Process ID: 800
```

```
Signal handling program started...
```

```
Program is running...
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```
Program is running...
```

```
SIGUSR1 received!  
User-defined signal received.  
Program is running...  
Program is running...  
Program is running...  
Program is running...
```

```
ananya@DESKTOP-42I5F93:~$ kill -SIGUSR1 800  
ananya@DESKTOP-42I5F93:~$ ./signal  
Process ID: 872  
Signal handling program started...  
Program is running...  
Program is running...
```

```
SIGUSR1 received!  
User-defined signal received.  
Program is running...  
Program is running...  
Program is running...
```

```
ananya@DESKTOP-42I5F93:~$ kill -SIGUSR1 872  
ananya@DESKTOP-42I5F93:~$ kill -SIGTERM 872  
ananya@DESKTOP-42I5F93:~$ ./signal  
Process ID: 947  
Signal handling program started...  
Program is running...  
Program is running...
```